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Characteristics of the Newly Eligible Population under Two Recent Updates of Lung Cancer Screening Recommendations

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Abstract

The United States Preventive Services Task Force (USPSTF) updated its lung cancer screening (LCS) recommendation in 2021 and the American Cancer Society (ACS) updated its LCS guidelines in 2023. Each update expanded screening eligibility criteria; thus increasing the total number of individuals eligible for lung cancer screening. However, it is not clear whether different population subgroups benefit equally from the recent updates of LCS recommendations in terms of becoming newly eligible. We identified 85,395 individuals who were between 50 and 80 years old, smoked cigarettes formerly or currently, and did not have a history of lung cancer from the Behavioral Risk Factor Surveillance System surveys of 2022. We used descriptive analysis to illustrate the weighted proportions of newly screening eligible population among different subgroups. We also applied multivariable logistic regression models to estimate the odds ratios of being newly eligible for LCS under each LCS recommendation update in 2021 and 2023. For both LCS updates, individuals who were non-Hispanic White, male, and had chronic obstructive pulmonary disease were significantly more likely to become newly eligible for LCS. A significantly larger proportion of older-age individuals became newly eligible under the 2023 ACS guideline. Noticeably, both guideline updates substantially increased the population eligible for screening among males and older individuals, two groups experiencing the majority of lung cancer incidence and mortality. Results also indicate the screening eligibility criteria updates did not increase the odds ratio of being eligible among racial/ethnic minority and female subgroups.

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Introduction

Lung cancer remains the leading cause of cancer-related deaths in the United States, accounting for nearly 25% of all cancer fatalities.(1) The financial impact is substantial, with national expenditures for lung cancer care estimated to be reaching $23.8 billion in 2020.(2) Beyond medical costs, the disease also has a profound impact on patients and families, with a five-year survival rate of only 26.6% for advanced stages, contributing to considerable morbidity and poor quality of life.(3)

The National Lung Screening Trial (NLST) demonstrated the efficacy of low-dose computed tomography (LDCT) screening in reducing lung cancer deaths among asymptomatic high-risk individuals.(4) In response, the United States Preventive Services Task Force (USPSTF) included LDCT in its 2013 lung cancer screening (LCS) recommendations for individuals at high risk for lung cancer, defined as: (1) age 55 to 80 years; (2) at least a 30 pack-year smoking history; and (3) current smokers or persons who smoke cigarettes formerly but had quit within the past 15 years (i.e., years since quitting (YSQ) ≤ 15).(5) In 2021, the USPSTF updated its LCS recommendations and expanded the screening eligibility criteria by lowering the starting age to 50 and reducing the pack-year requirement to 20 while keeping the YSQ criterion unchanged.(6) In 2023, the American Cancer Society (ACS) updated its LCS guidelines with one major difference from the 2021 USPSTF recommendation --- the removal of YSQ from screening eligibility criteria. (7–10) This decision was driven by a collection of evidence (systematic review, modeling, and observational studies) suggesting that, compared to never-smokers, the risk of lung cancer remains high for former smokers with YSQ greater than 15.(7–10) Table 1 summarizes the LCS eligibility criteria in the 2013 and 2021 USPSTF recommendations, as well as the 2023 ACS guideline. In this paper, we examined the characteristics of populations who were newly eligible for LCS from each guideline update to determine the characteristics of the newly eligible group associated with each update.

Materials and Methods

We analyzed the Behavioral Risk Factor Surveillance System (BRFSS) of year 2022, an annual health-related nationwide telephone survey covering approximately 400,000 adults from all 50 states, the District of Columbia (DC), and the three US territories of Puerto Rico, the U.S. Virgin Islands, and Guam. [https://www.cdc.gov/brfss/about/index.htm] Information on respondents' socio-demographic characteristics, health status, and disease history was extracted from the core module. [https://www.cdc.gov/brfss/annual_data/2019/pdf/overview-2019-508.pdf] The BRFSS is the largest ongoing telephone health survey system in the world, conducted by the Centers for Disease Control and Prevention (CDC) in collaboration with state health departments. [https://www.cdc.gov/brfss/index.html?utm] This database collects state-level data on U.S. residents regarding their health-related risk behaviors, chronic health conditions, and use of preventive services. The strengths of BRFSS include its large, representative sample size, robust data collection methods, and ability to provide national as well as state-level estimates. These features make the BRFSS a powerful data source for public health surveillance and policymaking. Numerous studies have utilized the BRFSS to estimate cancer screening rates across states

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in the U.S., e.g., colorectal cancer,(11) breast cancer,(12) and cervical cancers,(13) and prostate cancer.(14) These studies underscore the critical role of the BRFSS database in tracking national as well as state-level screening behaviors and guiding cancer control efforts. This study was exempt from human subjects review by the Institutional Review Board at the Sam Houston State University for the use of de-identified data.

We identified 445,132 adult respondents in the BRFSS 2022 survey. We assessed whether a respondent met the screening eligibility criteria based on three LCS guidelines: (1) the 2013 USPSTF LCS recommendations, (2) the 2021 USPSTF updated LCS recommendations, and (3) the 2023 ACS LCS guidelines. “Pack-year” was calculated as (“the average number of cigarettes smoked per day” × “smoke years”) ÷ 20,(15) as a standard pack of cigarettes contains 20 cigarettes. “Smoke years” for persons who smoked cigarettes formerly was the difference between a respondent’s age when he/she first started smoking and the age when he/she last smoked. For persons who smoke cigarettes currently the difference between a respondent’s current age and the age when he/she first started smoking. For persons who smoked cigarettes formerly, YSQ was calculated as the difference between a respondent’s current age and the age when he/she last smoked. We excluded respondents with missing information on their smoking history (n=58,995). We also excluded respondents who were younger than 50-year-old or older than 80-year-old (n = 173,744), and respondents who were between 50 and 80 of age and were never smokers (n=126,750). We further excluded individuals who had a history of lung cancer (n=248). The final study cohort included 85,395 persons who smoked cigarettes formerly or currently, did not have lung cancer and aged 50–80 from 50 states, DC, and 3 US territories (Figure 1).

We first calculated the weighted distribution of the entire study cohort and the population subgroups who met the LCS eligibility criteria based on each of the above three LCS recommendations or guidelines. Second, we estimated the proportions of newly eligible individuals by socio-demographic and health characteristics under two scenarios: (1) individuals who were not eligible based on the 2013 USPSTF recommendations but became eligible after the USPSTF updated LCS recommendations in 2021; and (2) individuals who were not eligible per 2021 USPSTF recommendations but became eligible under the 2023 ACS guidelines. Finally, we applied multivariable logistic regression models to evaluate the factors associated with becoming newly eligible for each scenario described above. Factors examined in the multivariable analyses included: socio-demographic characteristics and health characteristics. Socio-demographics included age groups, insurance status, sex, race/ethnicity, income, education, whether having a personal doctor, and smoking status. Race/ethnicity in the BRFSS was self-reported and classified as: non-Hispanic White (NHW), non-Hispanic Black (NHB), Hispanics, and other races/ethnic groups. Variables characterizing health included self-reported general health status and the presence of chronic diseases, e.g., heart attack, coronary disease, stroke, asthma, non-melanoma skin cancer, COPD, depression, kidney disease, and diabetes.

All statistical tests were two-sided, and the statistical significance level was 0.05. All analyses were conducted in STATA Version 15.1 (Stata Corp., College Station, TX) and had accounted for the complex survey design. Specifically, we used the weights of the BRFSS survey to adjust for the different response rates among different population

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subgroups so as to generate nationally representative estimates. [https://www.cdc.gov/brfss/annual_data/2017/pdf/weighting-2017-508.pdf] The manuscript followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) [https://www.equator-network.org/reporting-guidelines/strobe/] reporting guideline for cohort studies.

Data Availability Statement

We used publicly available dataset Behavioral Risk Factor Surveillance System (BRFSS) to conduct our analysis. The data are available at https://www.cdc.gov/brfss/data_documentation/index.htm. (RRID: SCR_012974)

Results

Among the entire cohort of 85,395 (unweighted) respondents (representing 38,600,000 individuals in the US), Figure 1 shows that 17,903 were eligible for LCS based on the 2013 USPSTF recommendations; an additional of 11,236 (representing 5,360,706 individuals in the US) became newly eligible for LCS following the 2021 USPSTF recommendations, leaving 56,256 (representing 25,119,938 individuals in the US) remained ineligible. Of those who remained ineligible, an additional of 12,324 respondents (representing 4,980,187 individuals in the US) became newly eligible from the 2023 ACS guidelines which removed the YSQ criterion.

Supplementary Table S1 shows the weighted distribution of the sociodemographic and clinical characteristics of the entire study cohort and the group of respondents who were eligible for LCS under (1) the 2013 USPSTF recommendations, (2) the 2021 USPSTF recommendations, and (3) the 2023 ACS guidelines. Supplementary Table S2 shows the weighted proportion of individuals who were screening-eligible based on the 2013 and 2021 USPSTF recommendations, and under the ACS 2023 guidelines. Supplementary Table S2 also shows the weighted proportion of newly eligible population out of the entire study cohort based on 2021 USPSTS screening guideline compared to 2013 USPSTF guidelines, as well as the weighted proportion of newly eligible population out of the entire study cohort based on the 2023 ACS screening guideline compared to the 2021 USPSTF guideline. Among persons who smoke cigarettes currently or formerly and aged between 50 and 80 (i.e., the entire study cohort), 21.03% met the screening eligibility criteria based on 2013 USPSTF LCS recommendation. When the LCS recommendations updated from 2013 USPSTF to 2021 USPSTF, an additional 13.89% of the study cohort were newly eligible for LCS, representing a total of 34.92% of the entire study cohort were eligible for LCS. Under the 2023 ACS guideline, an additional 12.90% of the study cohort were eligibility for LCS, representing a total 47.82% of the entire study cohort were eligible for LCS. Subgroups categorized by sociodemographic and clinical characteristics shows that the proportions of individuals who were eligible and who were newly eligible for LCS differ from each guideline update. We summarized the most noticeable findings below.

By age groups, age subgroups between 55 and 69 (26.65% for 55–59, 27.88% for 60–64, and 26.47% for 65–69) had the largest weighted proportions of eligible population based on 2013 USPSTF recommendation. The largest increase in gaining eligibility (31.72%) was observed in the 50–54 age group for the USPSTF 2021 update whereas the largest

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gain (29.51%) from the ACS 2023 update was among the eldest age group (age 75–79). After the two eligibility criteria updates, the age groups between 65 and 79 became the largest weighted proportions of eligible population based on the ACS 2023 recommendation. (51.11% for 65–69, 53.64% for 70–74, and 54.16% for 75–59) By gender and race/ethnicity, male and non-Hispanic White consistently had the largest proportions of eligible population under the three eligibility criteria. With the removal of the YSQ criterion (the ACS 2023 update), a lower proportion of females (vs. males), non-Hispanic Blacks, Hispanics, and those from other racial/ethnic groups (vs. non-Hispanic White) became newly eligible for LCS.

By household income and education, individuals in income group of "< $ 25,000" and education group "Did not graduate High School" consistently had the largest proportions of eligible population under the three lung cancer screening eligibility criteria. Individuals in income group of "< $25,000" and education group "did not graduate high school" had the largest increase in the proportions of individuals who become newly eligible when the USPSTF recommendations updated in 2021. On the contrary, these two subgroups had the smallest increase in the proportions of becoming newly eligible under the 2023 ACS guideline update.

By insurance status, the proportions of newly eligible among the uninsured was higher than the insured subgroups when the guideline updated from the USPSTF 2013 to the USPSTF 2021, whereas the proportion of newly eligible was higher among the insured than the uninsured subgroups when the guideline updated from the USPSTF 2021 to ACS 2023. By smoking status, current everyday smoker consistently had the largest proportions of eligible population under the three lung cancer screening criteria. When the guideline updated from USPSTF 2013 to USPSTF 2021, the proportions of newly eligible increased more among the current everyday smokers than among the current someday smokers and former smokers. When the USPSTF 2021 guideline updated to the ACS 2023 guideline, the only subgroup had newly eligible population was the former smoker subgroup.

By self-reported health status, the proportions of the eligible population increased across all three lung cancer eligibility criteria as individuals' self-reported health status declined. A higher proportion of individuals reporting "fair or poor" health status gained eligibility from the USPSTF 2021 update, whereas no clear pattern was observed under the 2023 ACS guidelines. The proportions of newly eligible individuals for those who had no personal doctors was higher than the proportion among those with one or more personal doctors when the USPSTF guideline updated from 2013 to 2021, whereas the opposite pattern was observed when the USPSTF 2021 guideline updated to ACS 2023 guideline. Individuals who had each of the comorbidities had higher proportions of eligible population across all three lung cancer screening eligibility criteria than the population without such comorbidity.

Figure 2 illustrates the state-specific weighted proportions of the population eligible for lung cancer screening under three guidelines: Panel A shows eligibility based on the USPSTF 2013 guideline, Panel B reflects the USPSTF 2021 guideline, and Panel C represents the ACS 2023 guideline. Additionally, the figure highlights the state-specific weighted proportions of newly eligible individuals following the two guideline updates: Panel D

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compares the USPSTF 2013 to the USPSTF 2021 guidelines, and Panel E compares the USPSTF 2021 to the ACS 2023 guidelines, across the 50 U.S. states and Washington, DC. To evaluate these proportions, we applied the same weighted methodology used for other variables as in Supplementary Table S2. The denominator consisted of individuals who smoke cigarettes currently or formerly and aged between 50 and 80 resided in each of the states and Washington DC in the 2022 BRFSS. The numerators were weighted populations of eligible or newly eligible individuals for lung cancer screening. The proportions were categorized into quantiles for each panel, with white indicating the lowest quantile and darker blue representing the highest quantile of the weighted proportions for eligible or newly eligible populations.

Figure 2 reveals notable patterns in state-specific proportions of the eligible population under each lung cancer screening guidelines. States such as California, Colorado, Connecticut, District of Columbia, Hawaii, Maryland, New Jersey, New Mexico, New York, Utah, and Washington consistently remained in the lowest quantile of the proportion of eligible population across all three guidelines. In contrast, Arkansas, Kansas, Kentucky, Louisiana, Mississippi, Ohio, Oklahoma, Tennessee, West Virginia, and Wyoming consistently ranked in the highest quantile. The largest increases in the proportion of the newly eligible population when transitioning from the USPSTF 2013 to the USPSTF 2021 criteria were observed in Arkansas, Idaho, Indiana, Kentucky, Louisiana, Michigan, Mississippi, Nebraska, North Dakota, Ohio, South Dakota, and West Virginia. Similarly, the largest increases in newly eligible population from the USPSTF 2021 to the ACS 2023 criteria were reported in Arizona, Delaware, Florida, Iowa, Kentucky, Maine, Michigan, Missouri, New Hampshire, North Carolina, Oklahoma, Rhode Island, and Vermont. These variations reflect differences in state demographics [https://data.census.gov/table?q=POPULATION+BY+AGE+AND+STATE] as well as smoking prevalence and intensity, [https://worldpopulationreview.com/state-rankings/smoking-rates-by-state] highlighting the impact of these factors on lung cancer screening eligibility. Supplementary Table S3 presents the weighted proportions of the eligible population and the newly eligible population for each state and Washington, DC.

Figure 3A shows the odds ratio (OR) of being newly eligible for LCS based on the USPSTF 2021 recommendation among individuals who did not meet the eligibility criteria of the USPSTF 2013 recommendations in multivariable analysis. The study sample corresponds to Study Cohort 1 in Figure 1. Compared to respondents in the 55–59 age group, the youngest group (age 50–54) were significantly more likely to be newly eligibly (OR=1.88, 95% CI: 1.62, 2.17) whereas the oldest age group were significantly less likely (OR= 0.44, 95% CI: 0.35, 0.55). The odds of becoming newly eligibility were significantly higher for individuals who had heart attack, COPD, and rated their general health as “good” and “fair and poor” (vs. “excellent and very good”). Conversely, respondents who were female, NHB, Hispanic, other race/ethnic groups (including Asian, American Indian/Alaskan Native, Native Hawaiian or other Pacific Islander, and Multiracial) (vs. NHW) and who had asthma (vs. no asthma) were significantly less likely to become newly eligible.

Figure 3B depicts the OR of being newly eligible for LCS under the ACS 2023 guidelines among those who did not meet the eligibility criteria in the USPSTF 2021 recommendation

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(Study Cohort 2 in Figure 1). The most noticeable pattern was a higher OR of being newly eligible in all age groups older than age group 55–59. Similar to the previous update in 2021 from the USPSTF, respondent who were female, NHB, Hispanics, and other races had significantly lower odds of being newly eligible. Those with a general health good, general health fair and poor, who had COPD, kidney disease, and diabetes had significantly higher odds of becoming newly eligible.

Discussion

Our study examined how two recent updates in LCS recommendations or guidelines affected LCS eligibility for persons who smoke cigarettes formerly and currently between the age of 50 and 80. We analyzed 2022 BRFSS data to examine the characteristics of individuals who became eligible for LCS as a result of two changes in the screening eligibility criteria: (1) from the USPSTF 2013 to the USPSTF 2021 recommendation update; and (2) from the USPSTF 2021 recommendations to the ACS 2023 guideline.

A recent analysis of the BRFSS 2019 survey found that among those who self-reported receiving CT for LCS, only about 20% met all three eligibility criteria (age, pack-year smoking history, and YSQ) based on the 2013 USPSTF recommendations and the percentage rose to 31% under the 2021 USPSTF recommendations.(16) Screening patterns observed in the real-world that deviate from the USPSTF recommendations could reflect physicians' decision that accounts for patients' personal risk for lung cancer,(17) and/or account for patients' preference when engaging in shared decision-making discussions about screening.(6) In addition, reasons such as inaccurate recorded smoking information in medical record may create barriers for providers to precisely identify potential eligible patients for LCS; this may subsequently lead to low screening rate.(18)

The USPSTF 2021 recommendations expanded the population eligible for LCS.(19–21) Yell et al. demonstrated that compared to the USPSTF 2013 guidelines, the screening eligibility changes in the USPSTF 2021 updated guidelines expanded the overall eligible population, notably increasing the representation of subgroups such as females, Black and Hispanic individuals, and seniors.(22) Similar to the USPSTF LCS recommendations updated in 2021,(19–21), the ACS 2023 LCS guideline also expanded the pool of eligible population. In addition, by removing the YSQ criterion, the ACS 2023 guideline is easier to implement in clinical settings. However, the change in individuals who became newly eligible for LCS differs between each update and across sociodemographic disease subgroups. The most noticeable difference is the differential gain by age subgroups. Specifically, the USPSTF 2021 LCS recommendations, by lowering the starting age of screening from 55 to 50, substantially increased the proportion of eligible individuals in the younger age group, whereas larger gain was found in the oldest age group (age 75–79) under the ACS 2023 LCS guidelines. This latter gain is likely due to many former smokers meeting the new ACS guidelines who previously were not eligible because they quit smoking more than 15 years ago.

Over the past 40 years, the lung cancer diagnosis rate for young and middle-aged women has increased faster than for men in the same age group, particularly among non-smokers.(23)

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Historically, men had higher incidence rates of lung cancer than women, but this trend has reversed in some racial and ethnic subgroups.(24) Our study indicated that despite updates to screening guidelines, the increase in screening eligibility for women has been lower or at most comparable to that for men. Women still represent a smaller proportion of the screening-eligible population overall. Further research should investigate changes in lung cancer screening eligibility rates among different age and sex groups to better understand the discrepancy between lung cancer diagnosis patterns and screening eligibility.

Findings from our study suggested that the updated guidelines may not reduce the differences in lung cancer screening eligibility rates among various racial/ethnic groups,(25–28) although the updates did increase the total numbers of LCS eligible individuals for all racial/ethnic subgroups. Previous research has also indicated that Black men are more likely to have lung cancer than White men;(29) thus, expanding LCS eligibility has the potential to reduce racial disparities in lung cancer mortality by increasing the number of Black males screened for lung cancer. However, like findings of other research,(30–32) we found that among those who were previously ineligible for LCS, non-Hispanic Black individuals had significantly lower odds of being newly eligible than non-Hispanic White individuals for both updated guidelines. The relatively smaller gain in screening eligibility among racial/ethnic minorities can in part be explained by three observations. First, non-Hispanic Black and Hispanic population who smoke cigarettes tended to be light smokers (33) and therefore were less likely to become newly eligible when pack-year was reduced from 30 to 20 under the USPSTF 2021 recommendation. Second, non-Hispanic Black and Hispanic who smoke cigarettes were less likely to quit smoking.(34, 35) And third, non-Hispanic White who smoke cigarettes were more likely to quit smoking at a younger age.(36) Therefore, when the ACS 2023 guideline removed the YSQ criterion, non-Hispanic White had a higher likelihood of being newly eligible. Another explanation is the racial/ethnic composition by age groups. A large number of newly eligible individuals under the ACS 2023 guideline were in the older age groups, which have a disproportionally higher percentage of non-Hispanic White.(37)

The proportion of screening-eligible individuals undergoing LDCT scans for lung cancer detection remains low, at approximately 16%.(16) Pinsky et al. estimated that based on findings from the National Lung Screening Trial (NLST) approximately 320 high-risk individuals would need to be screened with LDCT to prevent one lung cancer death over a 6.5-year period.(4, 38). Using our estimates in Supplementary Table S2, we conducted a similar back-of-the-envelope calculation to project the potential impact of expanding screening eligibility criteria under each guideline update. The transition from the USPSTF 2013 to the USPSTF 2021 screening recommendations would add approximate 5,361,540 (38,600,000*13.89%) individuals in the United States to become newly eligible for lung cancer screening. At the current screening uptake rate of 16%,(16) this corresponds to 857,846 (5,361,540*16%) individuals undergoing LDCT screening and is projected to prevent 2,681 (=857,846/320) lung cancer deaths among screening eligible population. Similarly, the update from the USPSTF 2021 recommendations to the ACS 2023 guidelines would add 4,979,400 (38,600,000*12.9%) newly eligible individuals. With the same screening uptake rate of 16%,(16) this translates to 796,704 individuals undergoing screening, potentially saving an additional 2,490 (796,704/320=2,490 lives among screening

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eligible population. Notably, the shift from the USPSTF 2013 to the USPSTF 2021 guidelines is projected to save 1,994 lives among individuals under 64 years of age, a relatively younger demographic. The further transition from the USPSTF 2021 to the ACS 2023 guidelines is projected to prevent an additional 634 deaths in this age group, but will have a substantially larger impact on the older groups (e.g., preventing 1,854 lung cancer deaths in the age groups above 65 years older)

In a systematic review, Sosa et al. discussed three potential sources of inequity in the LCS pipeline: screening eligibility, screening utilization, and post-screening behavior and follow-up care.(39) An important point emphasized in this review is that expanding screening eligibility alone does not guarantee increased screening uptake. For example, underinsured individuals may face difficulties accessing lung cancer screening resources, even if they are eligible for LCS. Targeted efforts to enhance lung cancer screening participation rates are essential. For instance, if the screening uptake rate were increased to 50%, the guideline expansion from 2013 to 2021 could prevent approximately 8,377 lung cancer deaths (38,600,000 * 13.89% * 50% / 320) among screening eligible population, while the transition from 2021 to 2023 could save an additional 7,780 (38,600,000 * 12.9% * 50% / 320) lives among screening eligible population. These findings highlight the critical importance of addressing barriers to lung cancer screening and implementing effective strategies to boost participation rates among eligible individuals, rather than focusing solely on expanding eligibility criteria.

Lastly, although the newest LCS guideline from the ACS 2023 guideline removed the YSQ criterion, it does not incorporate lung cancer risk factors unrelated to smoking, such as family history, lifestyles, or exposure to environmental toxins. These factors, while not easy to collect, have been incorporated in personalized lung cancer risk assessment tools.(40) Future updates to LCS guidelines should expand eligibility criteria beyond smoking history to include additional non-smoking related risk factors.

The BRFSS 2022 survey included data from all 50 states and the District of Columbia, which allows us to generate a comprehensive representation of lung cancer screening eligibility and trends in newly eligible individuals across every state in the United States. However, the dataset has certain limitations. First, the BRFSS relied on self-reported data on personal smoking habits and health status; thus, may be prone to recall bias or social desirability bias. Second, there were missing values for smoking-related variables. Additionally, since BRFSS data were collected through telephone or cellphone surveys, there was a potential for selection bias as individuals without access to phones or those less likely to participate telephone surveys may be underrepresented.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Prevention Relevance Statement

This study examines the evolving lung cancer screening guidelines and their public health impact. Analyzing the 2021 USPSTF and 2023 ACS updates, we show expanded eligibility but persistent disparities among sociodemographic groups. Our findings highlight the need for targeted interventions to improve screening uptake and promote equitable, inclusive prevention strategies.

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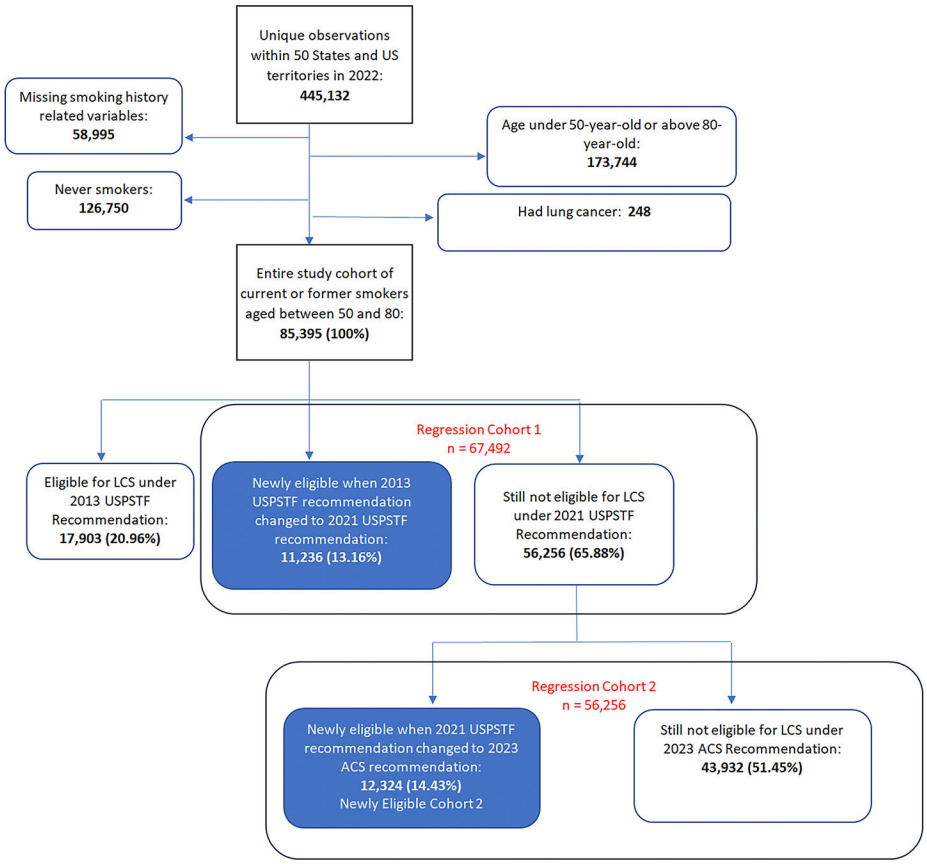


Figure 1.

Cohort Ascertainment and Change in Eligible Populations by Source of LCS Recommendations. Illustrated how we established our study cohorts.

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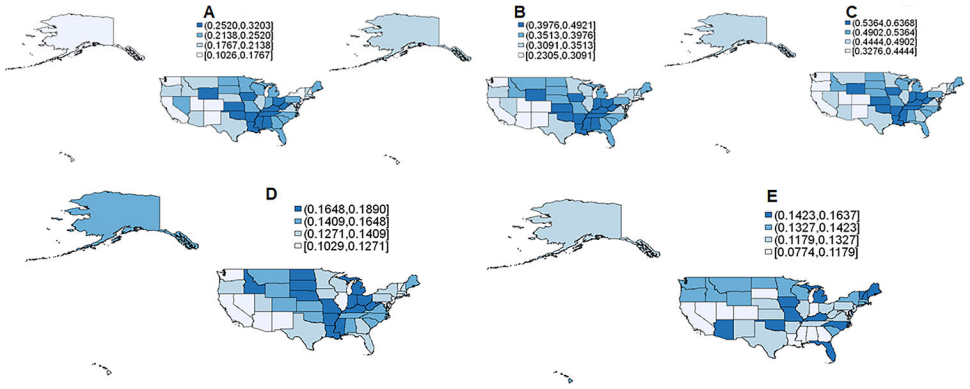


Figure 2:

Proportions of screening eligible population under three guidelines separately (2013 USPSTF, 2021 USPSTF, and 2023 ACS), and the proportions of newly eligible population across 50 states and District of Columbia of United States. Illustrated the proportions of screening eligible population and the proportions of newly eligible population across different states of United States. Panel A shows the proportions of eligible population under the 2013 USPSTF lung cancer screening guideline. Panel B shows the proportions of eligible population under the 2021 USPSTF lung cancer screening guideline. Panel C shows the proportions of eligible population under the 2023 ACS lung cancer screening guideline. Panel D shows the proportions of newly eligible population when 2013 USPSTF guideline changed to the 2021 USPSTF guideline. Panel E shows the proportions of newly eligible population when the 2021 USPSTF guideline changed to the 2023 ACS lung cancer screening guideline.

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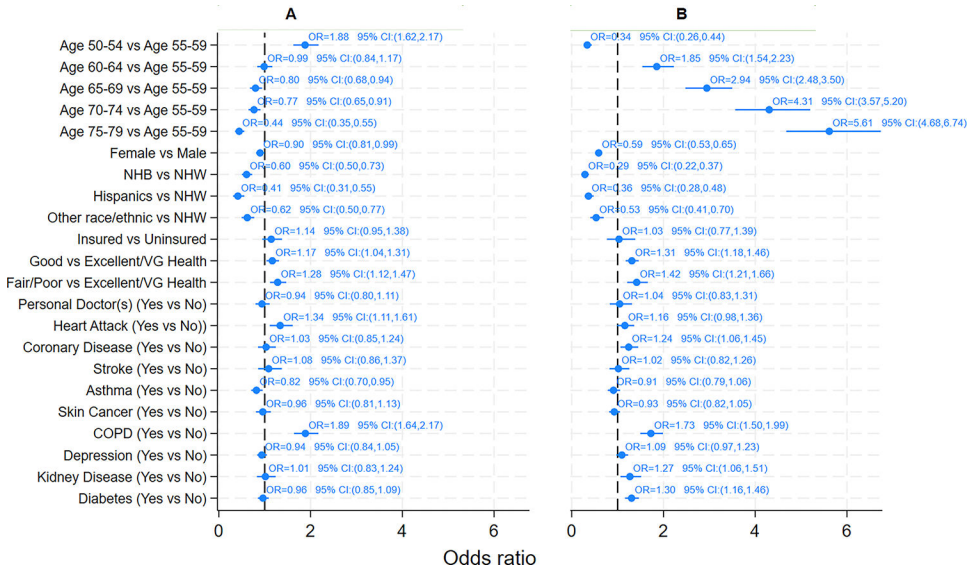


Figure 3:

Odds Ratio of Being Newly Eligible for LCS When Screening Guideline Updated. Illustrated the odds ratios of being newly eligible for LCS under two guideline updates: (1) From 2013 USPSTF to 2021 USPSTF; and (2) from 2021 USPSTF to 2023 ACS. Panel A illustrates the odds ratios of being newly eligible for LCS when the 2013 USPSTF changed to the 2021 USPSTF guideline. Panel B illustrates the odds ratios of being newly eligible for LCS when the 2021 USPSTF changed to the 2023 ACS guideline. Note: "Other Race/ethnic" included Asia, non-Hispanic American Indian/Alaskan Native, Native Hawaiian or other Pacific Islander, and Multiracial race/ethnic group. VG=Very Good. Other variables included in the logistic regression models but not shown as the variables were not statistically significant.

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Table 1:

LSC Screening Criteria Changes

	2013 USPSTF Recommendation	2021 USPSTF Recommendation	2023 ACS Recommendation
Criterion 1: Age	Age 55 to 80	Age 50 to 80	Age 50 to 80
Criterion 2: Pack-year	At least a 30 pack-year smoking history	At least a 20 pack-year smoking history	At least a 20 pack-year smoking history
Criterion 3: Smoking history	Current smokers or former smokers who had quit within the past 15 years (i.e., years since quitting (YSQ) ≤ 15)	Current smokers or former smokers who had quit within the past 15 years (i.e., years since quitting (YSQ) ≤ 15)	Not required